

Homework 02

ECE 473/573, Fall 2026

Due Date: 09/13 (Sun.) by the end of the day (Chicago time)

- You may use AI tools to solve the questions below. Here are a few hints you should follow:
 - Run the original code to understand what the issues look like.
 - Run the corrected code to make sure it actually works.
 - Even if the code works, explanation by AI tools is not necessarily true.
- 1. (1 point) A student implemented the function `appendRand` to append a random number to a slice as follows.

```
package main

import (
    "fmt"
    "math/rand"
)

func appendRand(a []float64) {
    a = append(a, rand.Float64())
}

func main() {
    a := make([]float64, 0)
    for i := 0; i < 10; i++ {
        appendRand(a)
        fmt.Printf("%d: %v\n", i, a)
    }
}
```

What is the expected output? What does the code actually output? Correct the code as needed and briefly explain why. If AI tools are used, ask for an alternative correction and explain which correction you would prefer to have.

2. (1 point) A student implemented a **Vertex** struct similar to what we have done in the lectures as below:

```
package main

import (
    "fmt"
)

type Vertex struct {
    X, Y int
}

func (v *Vertex) Move(dx, dy int) {
    v.X += dx
    v.Y += dy
}

func (v Vertex) String() string {
    return fmt.Sprintf("(%d,%d)", v.X, v.Y)
}

func main() {
    a := make([]Vertex, 0)
    for i := 0; i < 10; i++ {
        a = append(a, Vertex{X: i, Y: i * 2})
    }
    fmt.Printf("before move: %v\n", a)
    for _, v := range a {
        v.Move(1, 2)
    }
    fmt.Printf("after move: %v\n", a)
}
```

What is the expected output? What does the code actually output? Correct the code as needed and briefly explain why. If AI tools are used, ask for an alternative correction and explain which correction you would prefer to have.